

# FindTogether: AI-Powered Missing Person Detection and Collaboration System with Social Dashboard

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**Abstract**—Missing-person identification is a time-sensitive problem in which delays in analysing available evidence can significantly reduce the chances of successful recovery. The widespread use of surveillance cameras and mobile devices has resulted in large volumes of video data that may contain useful clues; however, manual inspection of this data is slow, inconsistent, and difficult under real-world conditions such as crowding, occlusion, poor lighting, and non-frontal views. This paper presents FindTogether, an AI-assisted platform for missing person detection and collaborative investigation that combines deep facial embeddings with clothing-based appearance descriptors and performs asynchronous video processing in a cloud environment. A YOLO-based person detector is used to identify individuals in video frames, after which facial and clothing features are extracted and fused into a unified similarity score to generate ranked candidate matches. An investigator dashboard supports human verification of AI-generated results, enabling efficient and transparent decision-making during missing-person investigations.

**Index Terms**—Missing Person Detection, Face Recognition, Person Re-Identification, Multi-Cue Fusion, Surveillance Analytics, Computer Vision

## I. INTRODUCTION

The rapid expansion of visual surveillance infrastructure has significantly influenced modern public safety and investigative practices. Urban environments, transportation systems, educational campuses, and residential areas are increasingly monitored using CCTV systems, while individuals continuously generate additional visual data through smartphones and social media platforms. Although these sources provide valuable evidence for missing-person investigations, the large volume of video data creates a major challenge for manual analysis.

Missing-person recovery is highly time-sensitive, as the likelihood of successful identification decreases as time passes. Traditional investigation approaches rely on physical notices, phone-based communication, and manual review of video

footage collected from multiple agencies. These methods often lack centralized indexing, prioritization mechanisms, and efficient cross-referencing capabilities. Moreover, prolonged manual inspection of surveillance footage is difficult to sustain and prone to human error, particularly in crowded scenes or under poor lighting conditions.

From a technical perspective, automated face recognition has become an important tool for identity verification. Deep learning-based embedding models such as FaceNet and ArcFace have demonstrated strong performance in controlled environments. However, real-world surveillance footage rarely provides ideal conditions. Faces are frequently partially occluded by masks, scarves, or hair, captured from non-frontal angles, or affected by motion blur and low resolution. Under such conditions, face-only recognition systems often experience reduced reliability.

Recent research in person re-identification has explored the use of full-body appearance information, including clothing color, texture, and spatial patterns, as an additional identity cue. Clothing features remain visible even when facial information is limited, although clothing alone cannot uniquely identify individuals because similar outfits may be worn by multiple people. Combining facial features with clothing-based appearance descriptors therefore provide a more robust identification strategy.

In this work, we present *FindTogether*, a multi-cue artificial intelligence system designed to support large-scale missing person detection. The system integrates deep facial embeddings with clothing-based appearance features and performs similarity matching across asynchronously processed video frames. A cloud-based backend manages data upload, frame extraction, feature computation, and similarity evaluation, while an investigator dashboard enables transparent human verification of candidate matches. This design ensures that

artificial intelligence assists investigators without replacing human decision-making.

The primary objective of this paper is to develop and evaluate a multi-cue AI-based missing-person detection system that combines facial embeddings, clothing descriptors, and asynchronous video processing to improve identification speed, accuracy, and scalability in real-world surveillance environments.

### A. Contributions

The FindTogether platform contributes at both the algorithmic and system levels to the problem of missing-person identification. By combining multiple visual cues within a scalable cloud-based architecture, the proposed system connects advances in computer-vision research with practical investigative workflows. Unlike face-only identification systems, the proposed approach models the complementary strengths and limitations of facial and clothing-based features while supporting structured human verification and review of AI generated matches.

The main contributions of this work are summarized as follows:

- Design of a multi-cue missing-person detection framework that integrates face embeddings with clothing-based appearance descriptors to improve robustness under occlusion, crowding, and low-quality surveillance footage.
- Development of an asynchronous cloud-based video processing pipeline capable of handling large CCTV recordings and user-submitted videos without affecting system responsiveness.
- Implementation of a weighted similarity-fusion model that combines facial and appearance features to reduce false positives while maintaining strong recall performance.
- Creation of a human-in-the-loop investigation dashboard that enables ranked result inspection, verification, and auditability of AI-assisted decisions.
- Experimental evaluation of the effect of fusion-weight selection and frame-sampling strategies on detection accuracy and processing latency.

### B. Organisation of Paper

The remainder of this paper is organized as follows. Section II presents a review of related work in face recognition, person re-identification, and surveillance-based visual analytics. Section III describes the architecture and design of the proposed FindTogether system. Section IV explains the detection algorithms and similarity-matching methodology used in the system. Section V discusses the experimental evaluation and performance analysis. Finally, Section VI concludes the paper and outlines possible directions for future work.

## II. LITERATURE SURVEY

Early research in human detection relied primarily on handcrafted feature extraction techniques. One of the most influential approaches was the Histogram of Oriented Gradients (HOG) method for pedestrian detection, which demonstrated robustness to moderate illumination and shape variations but struggled in complex and crowded environments [1]. Similarly, Haar cascade classifiers enabled real-time face detection and were widely adopted in early surveillance systems, although their accuracy and adaptability were limited under varying environmental conditions [2].

The introduction of deep learning significantly improved object detection performance. Region-based convolutional neural networks, including R-CNN, Fast R-CNN, and Faster R-CNN, demonstrated substantial improvements in detection accuracy compared to handcrafted methods [3], [4]. Later, single-stage detectors such as YOLO and SSD enabled faster inference speeds, making them suitable for real-time surveillance and large-scale video analysis applications [5], [6].

Recent research has focused on improving detection robustness in challenging conditions such as occlusion, crowd density, and low-resolution imagery. Multi-scale feature learning techniques have been proposed to detect small or distant individuals more reliably, while attention-based mechanisms help models focus on discriminative regions of interest while suppressing background noise.

Face recognition systems have also evolved from geometric and template-based approaches to deep embedding methods. Models such as FaceNet and ArcFace learn compact feature representations in which vector distances correspond to identity similarity, enabling efficient large-scale matching [8], [9]. Despite these advances, face-only recognition systems remain sensitive to pose variations, occlusion, and poor image quality, which are common in surveillance scenarios.

To address these limitations, person re-identification research has explored the use of full-body appearance features, including clothing color, texture, and spatial configuration. Part-based and appearance-based models have demonstrated improved identity consistency across camera views and time intervals [10], [11]. Tracking-by-detection frameworks such as DeepSORT further improve reliability by maintaining identity associations across consecutive video frames [12].

Although significant progress has been made in detection, recognition, and tracking individually, many existing surveillance systems treat face recognition and person reidentification as separate processes. Integrated systems that combine facial and appearance cues within a scalable investigative workflow remain limited. The FindTogether platform addresses this gap by integrating face embeddings and clothing-based descriptors within an asynchronous processing architecture that supports human-in-the-loop verification and transparent investigation workflows.

FindTogether System Architecture and Data Flow

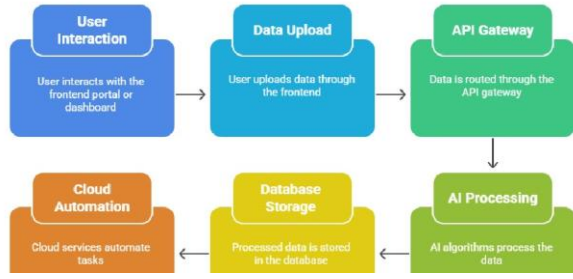


FIGURE 1: FindTogether system architecture and data flow

### III. PROPOSED SYSTEM

The FindTogether platform is designed as a scalable cloud based system that supports the complete workflow of a missing-person investigation, including case registration, evidence collection, automated matching, and investigator verification. The system follows a modular architecture in which data ingestion, AI processing, and user interaction are separated to improve reliability, scalability, and maintainability.

At a high level, the platform consists of five main components: a web-based interface for case registration and video uploads, a backend service layer responsible for authentication, storage, and job scheduling, an asynchronous video processing engine, AI inference modules for detection and feature extraction, and an investigator dashboard for reviewing results. This architecture ensures that user interactions remain responsive even when computationally intensive video analysis is being performed in the background.

When a new missing-person case is registered, reference images and descriptive information are stored in a centralized database. These images are processed to generate facial embeddings and clothing appearance descriptors, which together form the identity representation used during matching. Video evidence uploaded by users or collected from surveillance sources is stored securely in cloud storage and queued for background processing.

#### A. System Architecture

The data processing model used in FindTogether is similar to a distributed pipeline model. For instance, videos imported into the system are decomposed into individual frames depending on a predefined sampling rate before passing them through YOLO, an object detection system, to detect persons within the visual field. This software identifies all human figures in the scene. For each person, a bounding box is then created in order to separate the individual from the environment.

The cropped person images are then processed through two streams. One stream detects faces and derives a deep feature for each face based on a network that recognizes faces effectively. The second processing step calculates a feature based on clothing characteristics by examining color, texture, and layout

of each identified person. Therefore, a multi-cue feature for each person is a collection of two feature vectors.

All extracted features, together with corresponding information like frame number, timestamp, and video, will be archived in a search-engine able feature database. A similar matching service will match these potential representations against reference features linked to each active case of missing persons. The matching scores will then be ranked and made available to the investigator dashboard.

#### B. Processing Workflow

It is an event-driven system with a completely asynchronous workflow. On uploading a video, the backend service will create a processing job and push it into a task queue. The worker nodes will pull jobs from the queue and start to process frame extraction, person detection, and feature computation, respectively. Therefore, this solution is horizontally scalable, meaning more worker nodes can be added while increasing the amount of incoming data.

Upon extracting the features, the similarity-matching engine continues to match the candidate frames with the reference profiles of the registered missing persons. High-confidence matches are immediately surfaced through the dashboard to the investigators. In this sense, an incremental processing strategy may lead to the identification of potential leads even when the complete video has not been analysed, hence improving overall responsiveness and timeliness in investigations.

All events of the system activities, including video uploads and feature extraction and match evaluation events, are comprehensively logged to support auditability. Logging is a critical activity in investigative environments where transparency, traceability, and accountability are stringent requirements.

### IV. ALGORITHMS AND SIMILARITY FUSION

The FindTogether matching module is based on a multi-cue similarity model that combines facial and clothing appearance information. Let  $f_r$  and  $g_r$  denote the reference face and clothing feature vectors of a missing person, and let  $f_c$  and  $g_c$  denote the corresponding features extracted from a detected candidate in a video frame.

Facial similarity is computed using cosine similarity:

$$S_{face} = \frac{f_r \cdot f_c}{\|f_r\| \|f_c\|}$$

This measure evaluates the angular distance between embedding vectors and is widely used in deep face-recognition systems due to its stability and computational efficiency.

Clothing similarity is computed using a normalized distance metric:

$$S_{cloth} = 1 - \frac{\|g_r - g_c\|}{d_{max}}$$

where  $d_{max}$  represents the maximum possible distance in the clothing-feature space and is used to normalize the similarity value to the range  $[0,1]$ .

The final similarity score is obtained using weighted fusion:

$$S = \alpha S_{face} + (1 - \alpha) S_{cloth}$$

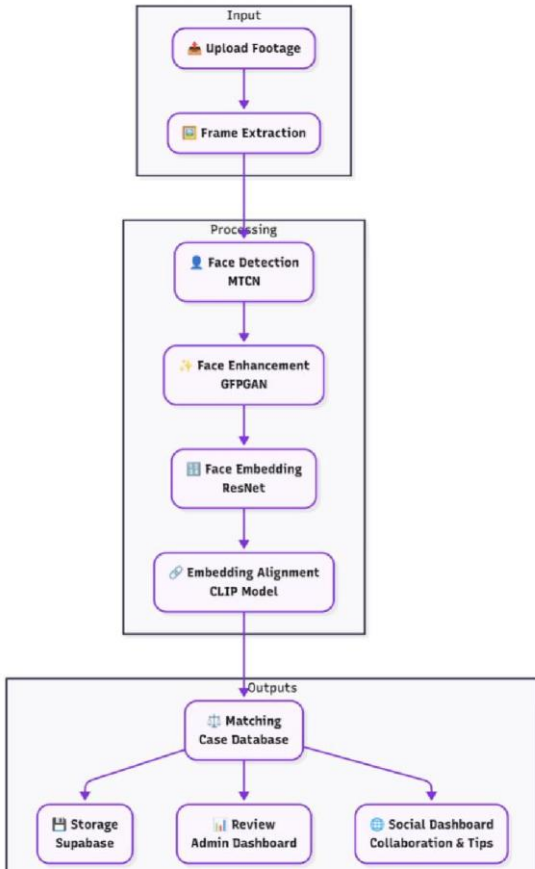


FIGURE 2: Level-0 data flow of the FindTogether platform

where  $\alpha \in [0,1]$  controls the contribution of facial and clothing cues. Larger values of  $\alpha$  prioritize facial similarity, while smaller values increase the influence of clothing-based appearance features. This adaptive fusion strategy improves robustness in surveillance conditions where facial visibility may be limited due to occlusion, distance, or lighting variations.

During experimentation, multiple values of  $\alpha$  were evaluated to determine the optimal balance between precision and recall. The system produces a ranked list of candidate frames for each missing-person case based on the fused similarity score  $S$ , allowing investigators to prioritize verification of the most probable matches.

## V. RESULTS AND EVALUATION

The FindTogether system was evaluated using surveillance style and mobile video scenarios designed to reflect real-world conditions such as varying illumination, camera distance, crowd density, and partial occlusion. The objective of the evaluation was to measure the system's ability to identify a missing person across different visual environments and to analyse the effectiveness of multi-cue similarity fusion.

Three matching configurations were tested: face-only similarity, clothing-only similarity, and fused similarity using the weighted model described in Section IV. In the face-only

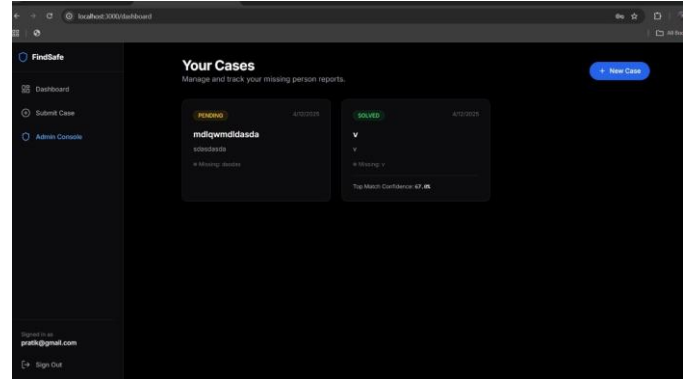


FIGURE 3: FindTogether investigator dashboard showing solved and pending cases

configuration, the system relied entirely on facial embeddings. Although this approach performed well when faces were clearly visible, performance declined in scenarios involving occlusion, non-frontal views, or low-resolution footage.

Clothing-only matching showed improved robustness in partially occluded situations but introduced false positives when multiple individuals wore visually similar clothing. These observations confirm that relying on a single visual cue is insufficient for reliable identification in realistic surveillance conditions.

The fused similarity model produced the most consistent results across all tested scenarios. By combining facial and clothing features, the system-maintained identification accuracy even when one cue was unreliable or unavailable. Experimental tuning of the fusion parameter  $\alpha$  showed that values between 0.7 and 0.8 provided the best balance between precision and recall, indicating that facial embeddings should remain the dominant cue while clothing descriptors provide complementary support.

Additional experiments evaluated different frame-sampling rates to understand the trade-off between computational efficiency and detection coverage. Lower sampling rates reduced processing time but occasionally missed brief appearances of the target individual. Higher sampling rates improved recall but increased computational cost. A moderate sampling rate was found to provide an effective balance between detection reliability and processing efficiency.

### A. Qualitative Analysis

Qualitative observations confirm that the system is capable of correctly identifying the target individual across multiple visual conditions. The experimental images show that the proposed approach successfully matches the same person across different frames and viewpoints.

In crowded scenes containing visually similar individuals, the system was able to correctly localize the target person using



the combined facial and clothing features. In low-light scenarios and distant camera views, where facial information was limited, clothing-based descriptors provided useful complementary cues for identification.



FIGURE 4: Detected missing person with bounding box and similarity score



FIGURE 5: Matching of the same individual across multiple camera views

### B. System Performance

The asynchronous processing architecture allows the system to handle large video uploads without interrupting user interaction. Although video analysis is computationally intensive, the use of background worker processes ensures that case registration, uploads, and dashboard interactions remain responsive.

This design makes the system suitable for real investigative environments where multiple users may interact with the platform simultaneously.

## VI. CONCLUSION AND FUTURE WORK

This paper presented FindTogether, an AI-based multi-cue missing-person detection system that integrates facial embeddings, clothing appearance descriptors, and asynchronous video processing. By combining complementary visual cues within a scalable cloud-based architecture, the system addresses common challenges in surveillance data, including occlusion, low resolution, and crowded scenes.

Experimental results indicate that the multi-cue fusion strategy improves identification robustness and reduces false positives compared to face-only or clothing-only approaches. The asynchronous processing pipeline further enables efficient analysis of large volumes of video data, making the system practical for real-world deployment.

Future work will focus on extending the system to support real-time CCTV stream processing, cross-camera tracking, and temporal behaviour analysis. Additional research will explore fairness, bias mitigation, and privacy-preserving feature extraction to ensure responsible deployment. Integration with law enforcement information systems and mobile applications for field investigators is also planned.

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